

Table 2: Various IPCop features

No.	Function	Description
1	Internet connectivity	Static IP, dynamic IP, PPPoE, modem
2	IPSec VPN	Enables two IPCops or one IPCop and second IPSec VPN device to establish site-to-site VPN by creating a tunnel in between.
3	Update	Security patch updates for IPCop
4	Backup	IPCop configuration backup
5	Various graphs	Details system, traffic and proxy graphs for ready reference
6	Proxy	Proxy server accepts requests from the Internal network, sends it out on the Internet, posting the received reply back to the corresponding internal system.
7	DHCP server	To allocate IP addresses on the internal network
8	Time server	To synchronise time on the internal network
9	Traffic shaping	Controls upload/download speeds by prioritising various services based on their ports
10	Port forwarding	Maps computer systems in the DMZ (or in the internal network); provides various services such as the email server, http server, ERP server, etc, to be accessible from the Internet
11	Proxy logs	Tracking the Internet access requests
12	System logs	Powerful logs for debugging IPCop

Table 3: IPCop add-ons

No.	Add-on name	Function
1	Advanced proxy	Enhances IPCop proxy capability: ▪ Adding local, radius and Windows authentication menu ▪ Cache management ▪ Web access control by the IP/MAC address ▪ Time-based access restrictions ▪ Blocking of unauthorised browsers/client software
2	URL filter	Blocks unwanted domains ▪ Blocking of websites based on categories ▪ Scheduled automatic blacklist updates
3	Update accelerator	An interesting add-on that caches various operating systems and anti-virus updates downloaded once, locally. Subsequent requests for the same files are supplied from the local cache, thus reducing bandwidth utilisation.
4	Calamaris report generator	Proxy report generator allows reports to be generated on various parameters such as: ▪ Request method ▪ Incoming / outgoing requests ▪ Protocols ▪ Originating IP address
5	Blocking outgoing traffic	This is one of the most important IPCop add-ons. By default, IPCop does not block requests from the internal network towards the Internet. Installing and enabling this add-on blocks all the outgoing requests towards the Internet. A granular control is established by allowing traffic based on: ▪ The incoming IP or MAC address ▪ Outgoing IP address ▪ Outgoing service
6	Zerina	Adds OpenVPN server functionality to IPCop. OpenVPN clients can download the generated certificate by the IPCop OpenVPN add-on to establish VPN connection.